A

Mini Project Report on

SMART GARBAGE MONITORING SYSTEM

Submitted to

Jawaharlal Nehru Technological University, Hyderabad

For the partial fulfilment of requirements for the award of the degree in

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

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(Autonomous, NBA (CSE, ECE, EEE) and NAAC 'A' Grade)

(Approved by AICTE, New Delhi, Affiliated to JNTUH, Hyderabad)

Nustulapur, Karimnagar 505481, Telangana, India

2024-2025

CERTIFICATE

This is to certify that the Mini Project Report entitled **"SMART GARBAGE MONITORING SYSTEM USING IOT" is** being submitted by MODUGU NANDEESHWARI (21271A0589), in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology** in **Computer Science & Engineering** to the **Jyothishmathi Institute of Technology & Science,** Karimnagar, during academic year 2024-2025, is a bonafide work carried out by them under my guidance and supervision.

The results presented in this Project Work have been verified and are found to be satisfactory. The results embodied in this Project Work have not been submitted to any other University for the award of any other degree or diploma.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to our advisor**, G. PRANAVI**, whose knowledge and guidance has motivated me to achieve goals I never thought possible. The time I have spent working under her/his supervision has truly been a pleasure.

The experience from this kind of work is great and will be useful to us in future. I thank **DR. R. JEGADEESAN**, Professor & HOD CSE Dept for his effort, kind cooperation, guidance and encouraging us to do this work and also for providing the facilities to carry out this work.

It is a great pleasure to convey our thanks to **DR. T. ANIL KUMAR**, Principal, Jyothishmathi Institute of Technology & Science and the College Management for permitting us to undertake this project and providing excellent facilities to carry out our project work.

I thank all the **Faculty** members of the Department of Computer Science & Engineering for sharing their valuable knowledge with us I extend our thanks to the **Technical Staff** of the department for their valuable suggestions to technical problems Finally Special thanks to our parents for their support and encouragement throughout our life and this course Thanks to all our friends and well-wishers for their constant support.

DECLARATION

I hereby declare that the work which is being presented in this dissertation entitled, "**SMART GARBAGE MONITORING SYSTEM USING IOT**", submitted towards the partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering, Jyothishmathi Institute of Technology & Science**, Karimnagar is an authentic record of our own work carried out under the supervision of

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To the best of our knowledge and belief, this project bears no resemblance with any report submitted to JNTUH or any other University for the award of any degree or diploma.

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ABSTRACT

Smart Garbage Monitoring System using Node MCU and Blynk is an IoT-based project. It was developed specifically to help scavengers and residents in monitoring the level of garbage in the dustbin through Blynk application, which makes it easy to empty the dustbin and optimize the garbage collection time. This project uses ultrasonic sensor to measure the level of garbage in the dustbin and the data can be viewed through mobile application, Blynk.
It will also notify users through email once the garbage bin capacity is full.

Waste management is one of the primary problems that the world faces irrespective of the case of developed or developing country. The key issue in the waste management is that the garbage bin at public places gets overflowed well in advance before the commencement of the next cleaning process. It in turn leads to various hazards such as bad odour & ugliness to that place which may be the root cause for spread of various diseases. Traditional garbage collection systems often lead to inefficiencies, such as delayed pickups, overflowing bins, and increased environmental pollution. To address these issues, this project proposes a Smart Garbage Monitoring System utilizing the Internet of Things (IoT) to revolutionize waste management.

The core objective of the proposed project is to ensure the enhancement of IoT based applications for collecting and managing waste efficiently. The conventional method of manually monitoring garbage in waste bins is a complex, time-consuming procedure that requires more human work, time, and money, and is in no way compatible with modern technologies.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT OVERVIEW

One of the important areas of concern in our environment is garbage management, which affects the human health. In the 21st century garbage collection is somewhat improved, compared to 20th century. Garbage is separately collected as reusable and non-reusable garbage's. But the amount of technology improved nowadays is not much reflected in garbage management. So, collecting, detecting and managing garbage's using the latest technology is one of primary need for society. Less human effort and more automation should be the goal for any garbage monitoring system. This project concentrates on developing a smart garbage monitoring and controlling system using Internet of Things (IoT).

We are living in the era of Smart cities where everything is planned and systematic. The problem we are facing is the population, which is rising rapidly. In recent years, urban migration has skyrocketed. This has resulted in the rise of garbage waste everywhere. Dumping of garbage in public places creates a polluted environment in the neighbourhood. It could cause several serious diseases to the people living around. This will embarrass the evaluation of the affected area. To reduce waste and maintain good hygiene, we need a systematic approach to tackle the problem.

The traditional way of manually monitoring the wastes in waste bins is a complex, cumbersome process and utilizes more human effort, time, and cost which is not compatible with the present-day technologies in any way. We propose a solution to this waste problem which manages the garbage waste smartly. This research paper proposes an IoT-based smart system based on clean waste management that assesses the level of waste on dustbins through sensory systems. In this system, the microcontroller is used as a visual connector connecting the sensor and the IoT system. This is an advanced method in which waste management is automated. This project IoT Garbage Monitoring system is a very innovative system that will help to keep the cities clean. This system monitors the garbage bins and informs about the level of garbage collected in the garbage bins via a web page. This web page also sends all information to garbage collection vehicles.

1.2 PROJECT PURPOSE

A Smart Garbage Monitoring System using IoT (Internet of Things) addresses the persistent challenges of waste management in modern urban environments. With the increase in urbanization and population, effective waste management has become a critical issue for cities around the world. This system leverages IoT technology to streamline the collection, monitoring, and management of waste, resulting in a cleaner environment, reduced costs, and enhanced operational efficiency.

The purpose of a Smart Garbage Monitoring System using IoT is to enhance waste management processes by leveraging technology to monitor and optimize garbage collection in real-time. Traditional waste management practices are often inefficient, leading to issues such as overflowing bins, unhygienic conditions, and increased operational costs. By integrating IoT (Internet of Things) technology into waste management, this system aims to address these challenges through automation, data-driven decision-making, and improved resource utilization.

- **Efficient Waste Collection**: A smart garbage monitoring system uses IoT-enabled sensors installed in waste bins to measure the fill levels. These sensors send real-time data to a centralized platform, helping waste management authorities know when bins are nearing capacity. This minimizes unnecessary trips to empty bins that are not full, saving fuel, time, and labour.
- **Reduction of Overflowing Bins**: Overflowing bins are a common issue in urban areas, leading to environmental pollution and public health risks. By providing timely alerts when bins reach their maximum capacity, the system ensures that garbage is collected before it spills over, maintaining cleanliness and hygiene.
- **Cost Optimization**: With data-driven route planning based on bin fill levels, garbage collection trucks can optimize their routes to cover only the bins that need attention. This reduces operational costs, such as fuel consumption and vehicle wear and tear, while also cutting down on manpower requirements.
- **Environmental Benefits**: By reducing unnecessary garbage truck trips and ensuring timely waste collection, the system contributes to lower greenhouse gas emissions. Additionally,

proper waste management prevents littering and contamination of surrounding areas, contributing to a cleaner and greener environment.

- **Public Health and Hygiene**: Proper waste management prevents the spread of diseases caused by accumulated garbage, pests, and bacteria. A smart system ensures that waste is removed promptly, maintaining healthier living conditions for communities.
- **Automation and Scalability**: IoT technology allows for automation of the monitoring and reporting processes, reducing human intervention and errors. The system is scalable, making it adaptable to cities of varying sizes, from small towns to large metropolitan areas.

1.3 PROJECT SCOPE

A smart garbage monitoring system using IoT provides an innovative solution to waste management challenges in urban and rural areas. The scope of such a system encompasses several key areas:

- **Real-time Monitoring and Alerting:**
- **Sensor Integration:** Integrating sensors like ultrasonic or weight sensors into garbage bins to continuously monitor fill levels.
- **Data Collection:** Collecting real-time data on bin fullness, location, and environmental factors (temperature, humidity).
- **Alert Generation:** Triggering alerts or notifications to relevant authorities when bins reach a predefined fill level or other critical conditions.
- **Optimized Waste Collection Routes:**
- **Route Planning:** Utilizing collected data to optimize waste collection routes, minimizing travel distances and reducing fuel consumption.
- **Dynamic Scheduling:** Adjusting collection schedules based on real-time bin fill levels and traffic conditions.
- **Predictive Maintenance:**

- **Sensor Health Monitoring:** Tracking the health and performance of sensors to identify potential issues and schedule maintenance.
- **Predictive Analytics:** Analysing historical data to predict sensor failures and plan preventive maintenance.

1.4 PROJECT FEATURES

- **Real-Time Monitoring**:
  IoT sensors installed in garbage bins measure the fill level in real time, providing continuous updates on waste levels to a centralized system.
- **Automated Alerts**:
  The system sends notifications to authorities or waste management teams when a bin is near capacity, ensuring timely collection and preventing overflow.
- **Route Optimization**:
  Integration with GPS and navigation systems allows for optimized garbage collection routes, reducing fuel consumption, time, and operational costs.
- **Data Analytics**:
  The system collects and stores data on waste patterns, enabling analysis to predict peak waste generation periods and improve future planning.
- **Energy Efficiency**:
  Designed to operate with low-power IoT devices, the system is energy-efficient and can use renewable energy sources like solar panels for powering sensors and devices.
- **Mobile App**:
  A user-friendly interface allows stakeholders to monitor bin status, analyse waste trends, and access alerts. Citizens can also find the nearest available bins.

CHAPTER 2
LITERATURE REVIEW

The global increase in urban population and waste generation in areas necessitates the development of efficient waste management solutions. Traditional methods of waste collection and monitoring are often labour-intensive, time-consuming, and inefficient. IoT-enabled smart trash cans are used for waste and landfill. It can be used for waste management. This may be the lack of general and public grade IOT-based smart waste disposal devices. This is a technologically complex field at a time when all your data is stored in the cloud with access to high-speed data in real time, and its data relies on the trash of IoT management racks.

Solid waste is a low-level development problem, as evidenced by the volume of urban waste due to rapid development and economic development, using non-substituted quasi-static and dynamic waste storage methods. Smart cities are the next step in human settlement focused on the integration of the business and residential environment. The network of the future and the Internet of Things have changed the small devices that form the background of human life on earth, such as the smart dust system used to deal with the hard waste in the city. Smart garbage monitoring systems leveraging the Internet of Things (IoT) have emerged as a promising alternative to address these challenges. This literature review explores existing research and advancements in smart garbage monitoring systems.

RELATED WORKS:

TITLE 1: A Garbage Disposal Management System to Support Swachh Bharath Mission

AUTHOR: - Denicke Solomon H, Abishek R, Ramachandra T, et al

YEAR: - April 2022

Currently, people prefer to live in a clean environment to be healthy. Both organic and inorganic waste from homes is disposed of in community bins managed by local councils or municipalities or companies. Wastewater generated by stray dogs pulling their food out and people throwing garbage out of moving vehicles causes serious problems for neighbours. While the Indians are aware of the Swachh Bharath Mission, there are many challenges. The work presented in this paper is to facilitate waste management using Arduino, connected

network, two ultrasonic sensors, servo motors, GSM/GPRS module, 4 X 4 matrix keypad, piezo buzzer and LEDs. The waste management system is ideal for busy public areas such as universities, shopping malls, train stations and airports. While it's designed to cost more than regular trash, it lasts longer and is easier to dispose of.

Smart storage boxes are popular because they are easy to manage and user-friendly. The standard prohibits the public from throwing garbage on the trash can. It is also used to reduce waste. Going one step further and sending materials to local storage instead of the website will save a lot of time and also keep existing cleaners safe on the job. Investing in designs is inexpensive because they last longer without changing. Therefore, the waste management strategy will assist in the realization of the Swachh Bharath Mission concept.

TITLE 2: A Smart Automated Garbage Management System to Replace Human Labour

AUTHOR: - Souvik Mondal, Sayantani Das, Shirsendu Banerjee, Koushik Pal

YEAR: - April 2022

The main purpose of this project is to create an intelligent waste management system that uses a movable chassis and a robot arm placed on a two-shelf bin, sensors and some electronic equipment. Urban waste is known to employ large numbers of workers to sort and manage waste. This process of identifying waste is done by workers with their own hands, which puts them in direct contact with waste and poison. It is very dangerous and bad for their health and can be caused by some bacteria, viruses or methane, carbon dioxide etc. They will be affected by many deadly diseases caused by certain pollutants such as the robot arm of Model will first pick up the garbage, then sensors will recognize the material, determine whether it is biodegradable or non-biodegradable waste, and place it in the appropriate place. In other words, the system works in a way that makes human hands. Thus, the system ensures that workers are not directly exposed to harmful and toxic whips in addition to pollution. This will help them help the community learn about health.

In this concept, we present a brief description of the layout components and circuit connections. This is the perfect solution for handling large volumes of waste. The logic behind

the different waste is very simple because it is just our sensor design compared to other models applying machine learning or spectroscopy, so it's easy to design and use.

TITLE 3: - Deep Learning based Garbage Detection for Smart City Applications

AUTHOR: - Dr.K. Lakshmi, Iswarya.k, N. Sheik Abdullah, K.R. Monica Bharathi

YEAR: - April 2022

The implementation of smart cities continues across India. Smart cities focus on the work that needs to be done to take care of the management and efficient use of resources. An important task in city management is to keep the city clean, and employees are used to it. Waste Management Program will help manage the planning and distribution of waste products. This article presents an automated scum monitoring system that uses the Faster CNN method to monitor scum flow. - YOLOv3 is a real-time item detection set of rules that identifies precise gadgets in videos, stay feeds, or images. YOLO makes use of functions found out via way of means of a deep convolutional neural community to come across an item. YOLO in a one-step item detection approach in contrast to R-CNN and Faster R-CNN.

Title 4: Smart Waste Management System using IoT.

AUTHOR: - Abdulla Al Mamun, Et al

Year: - 2020

This article shows how to implement smart waste management using IoT. When the waste reaches the maximum level, the instructions ensure that the waste is collected as soon as possible. Thus, the system will provide accurate information that will increase the efficiency of the system. Monitoring waste in real time with the help of sensors and wireless communication will reduce all the trips required by GCVs, thus reducing all expenses related to garbage collection. As a result, trash cans will be thrown away when full, giving way to cleaner cities, better infrastructure, and a cleaner environment. Good idea for good waste management. The system combines information and algorithms for waste management processes. Box detection is achieved by combining multiple detection methods. Various experiments were conducted by the authors to evaluate the performance of the model, and the authors concluded that these results show that the proposal and algorithms are of good value for waste management.

CHAPTER 3
EXISTING & PROPOSED SYSTEM

3.1 EXISTING SYSTEM

The existing waste management systems across the world are varied and depend largely on regional infrastructure, technological capacity, public awareness, and governmental support. The garbage is collected by municipal people by weekly once or by 2 days once. By this it leads to the unhygienic condition and cause different diseases which spoils the environment [6-10]. The smell will be heavy and produces air pollution and spreads disease. So, there is an increasing need for holistic approaches to waste management that integrate sustainability, innovation, and community involvement, especially as the volume of waste continues to rise globally. The street dogs and animals eat the waste food and spreads over the area and creates dirty environment to avoid such situation we are planning to design IoT Based Garbage Management for Smart Cities.

3.2 EXISTING SYSTEM DISADVANTAGES

- Cleaning of bins is not done properly.
- It causes the unhygienic condition and spoil the environment.
- Bad smell spreads and may cause illness to human beings.
- More traffic and noise.

3.3 PROBLEM STATEMENT

There are millions of public boxes out there for people to use and are released by public authorities within days. Now the problem is that not all bins are filled for the same price and waste trucks waste time checking each base. This results in higher fuel consumption, performance, and costs.

It has also been observed that most people do not use the litter box because they have no interest in going near the litter box. The sensors will show the daily status of containers that are

often in poor condition and not cleaned, including garbage. A cloud server has also been set up where city officials can get information about unattended items and containers. Work is carried out in a relaxed manner; necessary things are done to clarify the care of the green environment.

Efficient waste management is a significant challenge for municipalities, institutions, and businesses due to increasing urbanization and population growth. Conventional garbage collection systems rely on fixed schedules and manual monitoring, leading to several inefficiencies. Additionally, the lack of real-time data on garbage levels makes it difficult for authorities to prioritize waste collection efforts, resulting in delays and unhygienic conditions. The challenge is to design and implement a smart garbage monitoring system using IoT (Internet of Things) technologies. This system should:

- Monitor the real-time fill level of garbage bins using sensors.
- Provide notifications to waste management authorities when bins are nearly full.
- Optimize waste collection routes based on bin fill levels, reducing fuel consumption and collection time.
- Enable scalability and integration with existing smart city frameworks.
- Encourage a cleaner and more sustainable environment through better waste management practices.

By leveraging IoT technologies, this solution aims to enhance operational efficiency, reduce environmental impact, and improve the quality of urban living.

3.4 PROPOSED SYSTEM

The existing system has the limitations as time consuming, trucks go and empty the containers, even they are empty. The cost is high with unhygienic environment. Even the bad odour causes the unhealthy environment. So, proposed model talks about how to make use of the recent advancements in technology to make our place clean and tidy. The implementation starts by setup ESP8266 by flashing the latest version of the firmware. This enables the Blynk libraries efficiently communicate and avoid producing error. To flash the latest firmware, download the ESP8266 flasher tool and the latest firmware from the internet which would be in the bin format and flash the ESP8266 with it. Once the ESP8266 flashing done, other components can be added to the configuration.

This proposed system assures the cleaning of dust bin soon when the garbage level reaches its maximum threshold. Thus, this system helps to keep the city clean by informing about the garbage levels of the bins by providing SMS to the respective person.

3.5 PROPOSED SYSTEM ADVANTAGES

- **Time Efficiency**:
  The system ensures that garbage containers are emptied only when they are full, eliminating unnecessary trips by garbage collection trucks to empty containers that are not full.
- **Cost Savings**:
  By reducing unnecessary trips for garbage collection, the system minimizes fuel and labour costs, resulting in significant cost savings for municipalities or waste management services.
- **Improved Hygiene**:
  Regular monitoring and timely emptying of containers reduce the chances of garbage overflow, thus maintaining a cleaner environment.
- **Odor Control**:
  The system helps to prevent the accumulation of waste for long periods, which can otherwise cause bad odours and create an unhealthy environment.
- **Environmentally Friendly**:
  By optimizing garbage collection schedules, the system reduces the carbon footprint of waste management operations through fewer truck movements and emissions.
- **Use of Advanced Technology**:
  Incorporating IoT technology allows real-time monitoring and data collection, improving the overall efficiency and responsiveness of the waste management system.
- **Scalability**:
  The system can be implemented in a variety of settings (urban, rural, industrial, etc.) and scaled as needed to accommodate additional garbage containers or zones

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

Hardware interfaces specify the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements.

- Node MCU (ESP8266)
- Ultrasonic Sensor (e.g., HC-SR04)
- USB Cable
- Jumper Wires
- Power Supply
- Garbage Bin

4.2 SOFTWARE REQUIREMENTS

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are the software requirements.

- **Arduino IDE:** To program the Node MCU microcontroller.
- **Blynk Application:** Mobile App
- **Programming Language:** C++

4.1 HARDWARE REQUIREMENTS DESCRIPTION:

4.1.1 Node MCU:

Node MCU is an open-source Lua based firmware and development board specially

targeted for IoT based Applications. It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. Node MCU Dev Kit has Arduino like Analog (i.e. A0) and Digital (D0-D8) pins on its board.

It supports serial communication protocols i.e. UART, SPI, I2C.

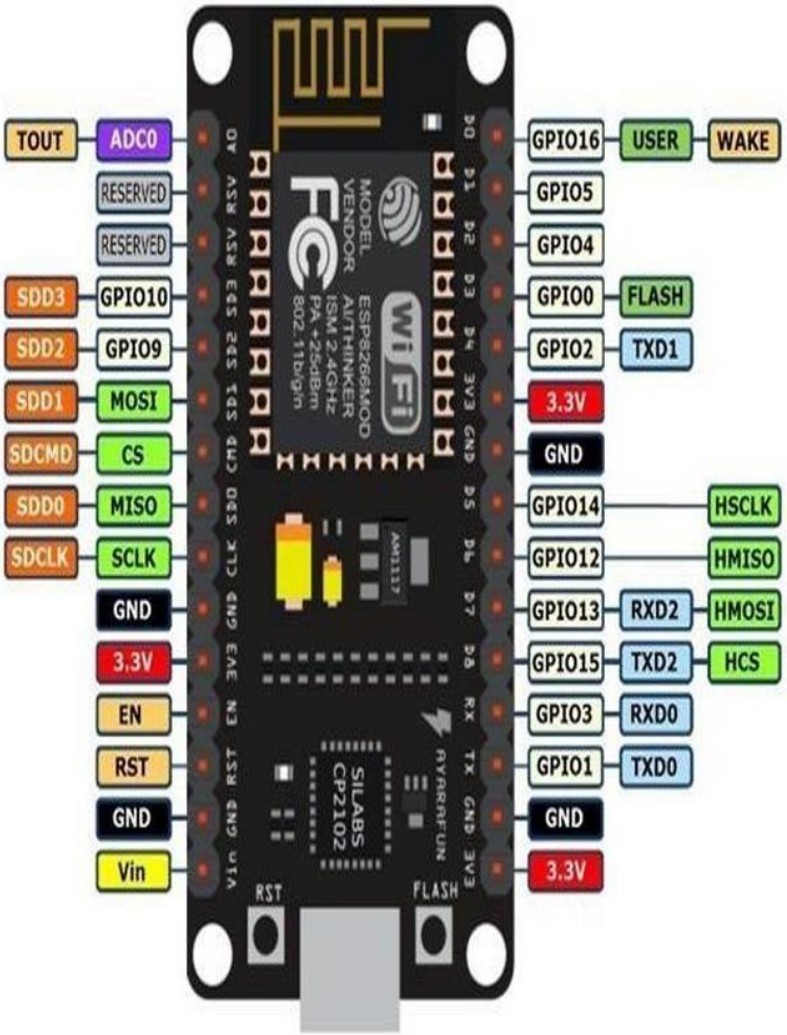

Fig 4.1.1: NODE MCU

D0 = GPIO16;

D1 = GPIO5;

D2 = GPIO4;

D3 = GPIO0;

D4 = GPIO2;

D5 = GPIO14;

D6 = GPIO12;

D7 = GPIO13;

D8 = GPIO15;

D9 = GPIO3;

D10 = GPIO1;

LED_BUILTIN = GPIO16 (auxiliary constant for the board LED, not a board pin);

4.1.2 ULTRASONIC SENSOR:

An ultrasonic sensor is an electronic device that can measure the distance of a target object by sending out ultrasonic sound waves, and converting the reflected sound into an electrical signal. Basically, such a sensor uses a transducer to send and get ultrasonic pulses that in turn send back information about an object's proximity.

Ultrasonic sensors have two key components: one is the transmitter that emits the sound using piezoelectric crystals and the other is the receiver that receives the sound after it has travelled to and from the target.

Fig 4.1.2: Ultrasonic Sensor

4.2 SOFTWARE REQUIREMENTS DESCRIPTION:

4.2.1 Arduino IDE:

Fig 4.2.1: Arduino

The Arduino Integrated Development Environment (IDE) is an open-source software application that allows users to write, compile, and upload code to Arduino boards and

compatible microcontrollers. It is widely used in electronics, IoT, and embedded systems development due to its simplicity, flexibility, and user-friendly interface.

Here are the steps to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: Visit the Official Arduino Website and Select Your Operating System.

Open your web browser and go to the https://www.arduino.cc/

Step 2: Download Arduino IDE Software.

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

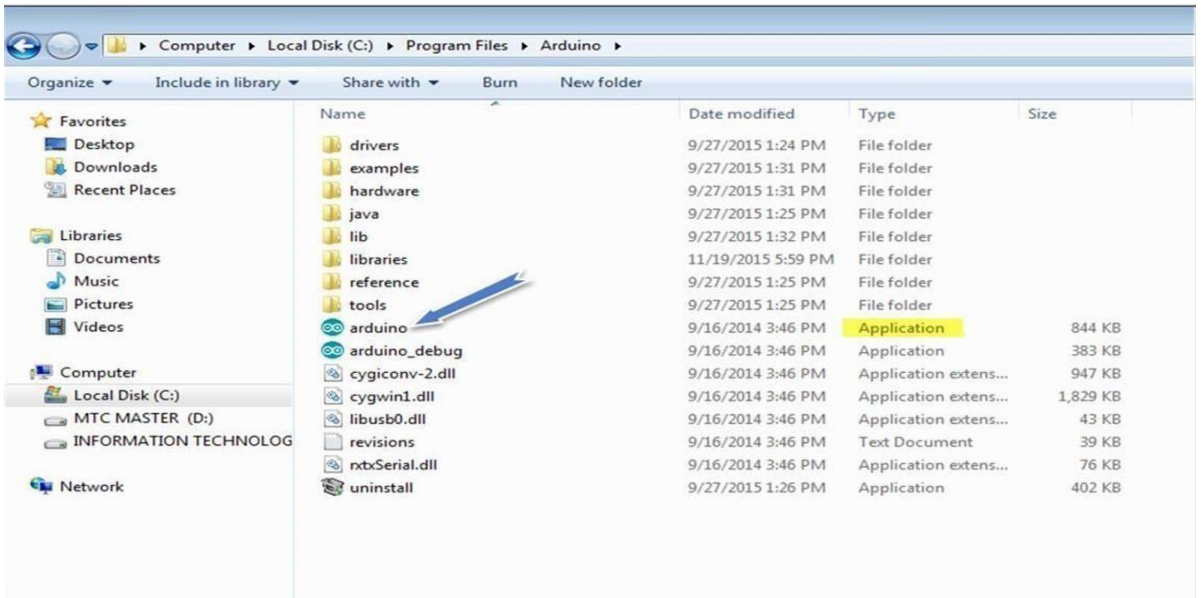

Fig 4.2.1(a): Downloading the Arduino IDE software

Fig 4.2.1(b): Unzipping the downloaded file

Step 3: Power up your board.

The Arduino Uno draw power from either, the USB connection to the computer or an extern al power supply. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port.

Connect the Arduino board to your computer using the USB cable. The green power LED (labelled PWR) should glow.

Step 4: Add ESP8266 Board Support.

The Node MCU board is based on the ESP8266 microcontroller, so you need to install support for ESP8266 boards**.**

Steps:

1. Open the Arduino IDE.
2. Go to **File > Preferences.**
3. In the **Additional Boards Manager URLs** field, add the following URL:

   https://arduino.esp8266.com/stable/package_esp8266com_index.json

4. Click OK.

Step 5: Install the ESP8266 Board Package 1.

Go to Tools > Board > Boards Manager.

2. In the search bar, type ESP8266.

3. Select esp8266 by ESP8266 Community and click Install.

4. 4.Wait for the installation to complete

Step 6: Select the Node MCU Board

1. Go to Tools > Board > ESP8266 Boards.

2. Select Node MCU 1.0 (ESP-12E Module)

Step 7: Configure the Port

1. Connect your Node MCU to your computer via USB.
2. Go to Tools > Port and select the COM port corresponding to your Node MCU.

3. On Windows, this will appear as COMx.

4. On macOS/Linux, it will appear as something like /dev/cu.SLAB_USB to UART or /dev/ttyUSBx.

Step 8: Configure Upload Settings

1. Flash Size: 4MB (32Mb)
2. CPU Frequency: 80 MHz
3. Upload Speed: 115200 baud (default) or 921600 baud

Step 8: Test the Configuration

1. Open an example sketch:

   Go to File > Examples > ESP8266 > Blink.

2. Upload the sketch to test if the configuration is correct.

4.2.2 BLYNK APPLICATION:

Fig 4.2.2: Blynk App

Blynk is the complete software package needed to model, deploy and manage connected electronic devices of any size, from individual IoT projects to millions of connected commercial products. With Blynk, anyone can connect their devices to the cloud and create code-free iOS, Android, and web apps to analyse real time and historical data from devices, manage them from anywhere in the world, receive important notifications and much more. This Blynk ESP8266 Smart Control Relay has the following functions: Control Home Appliances via Wi-Fi (Blynk IoT App). Manage your home appliances with the Blynk web dashboard. Control home appliances with switches or buttons. Monitor real-time feedback in the Blynk IoT App. Thus, you can easily complete a home automation project in your home using only Node MCU and relay modules. Or you can use a standard PCB design for this project. IoT technology has the potential to revolutionize business, but many companies struggle to use it effectively due to competition, lack of expertise, uncertain business models, and lack of experienced engineers. Whether you're starting a new business or an established company to accelerate growth, Blynk will help you break these barriers and create IoT solutions that solve your real business problems.

CHAPTER 5

PROJECT DESCRIPTION

The Smart Garbage Monitoring System is designed to automate waste management by monitoring the garbage levels in bins using IoT (Internet of Things) technology. This system leverages a **Node MCU (ESP8266)** microcontroller, an **ultrasonic sensor (HC-SR04)**, and the **Blynk mobile app** to provide real-time updates on waste levels, ensuring efficient waste collection and reducing operational costs.

CIRCUIT DIAGRAM:

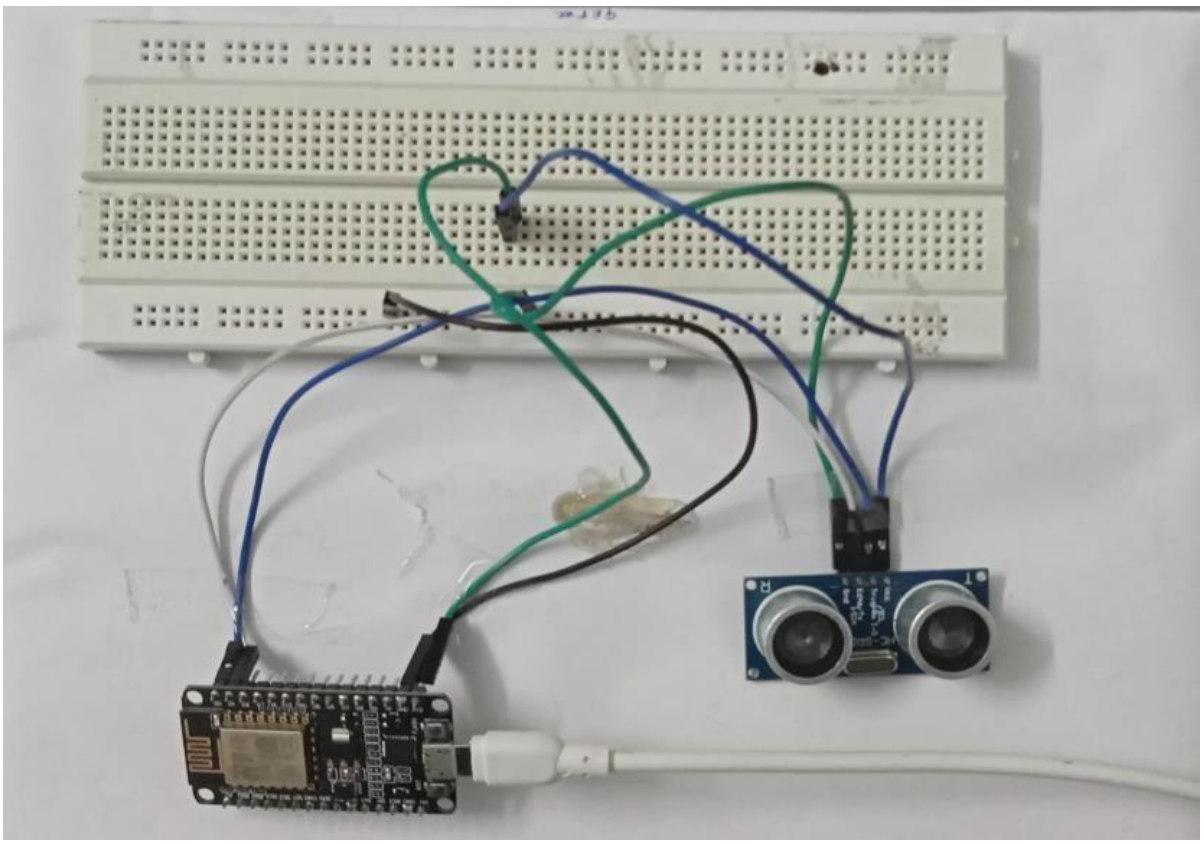

Fig 5.1: Circuit Diagram

BLYNK CONFIGURATION:

Setting Up the Blynk App

Step 1: Install Blynk Application from Google Play store.

Step 2: Create an account at Blynk Application. For new users Sign Up and old users just Log in.

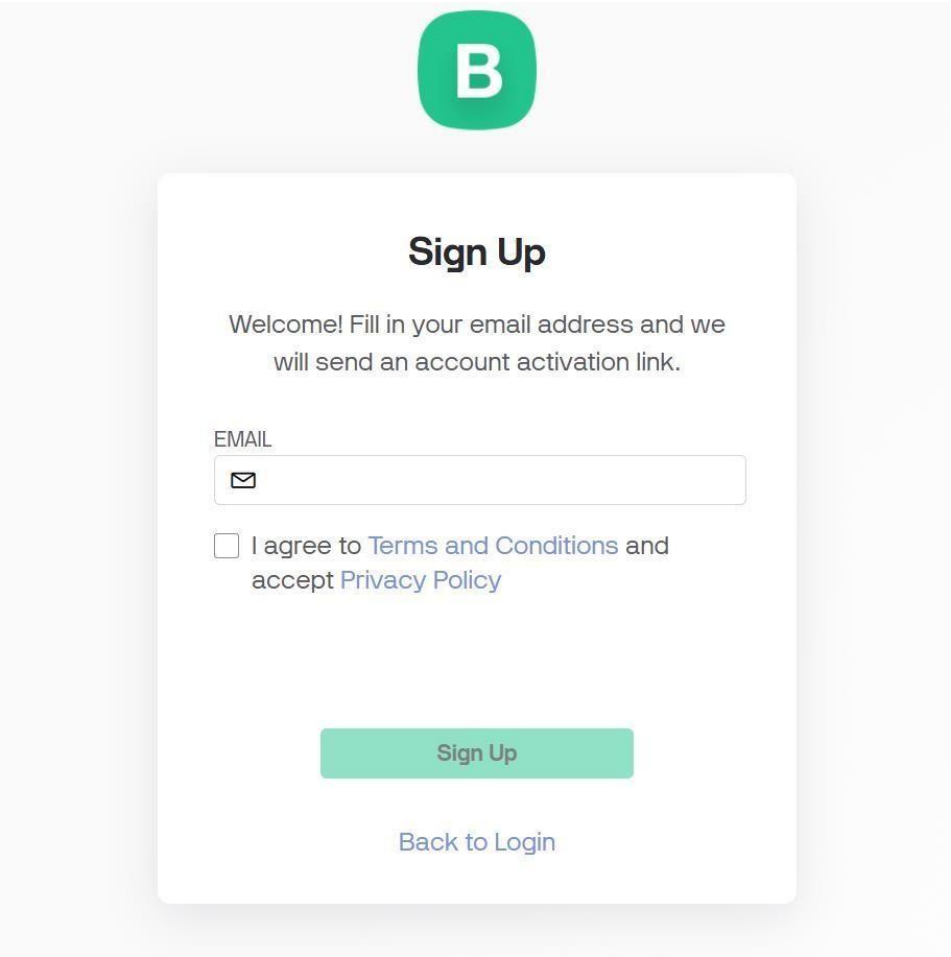

Fig 5.2: Blynk Sign Up

Step 3: Create a new template for ESP8266.

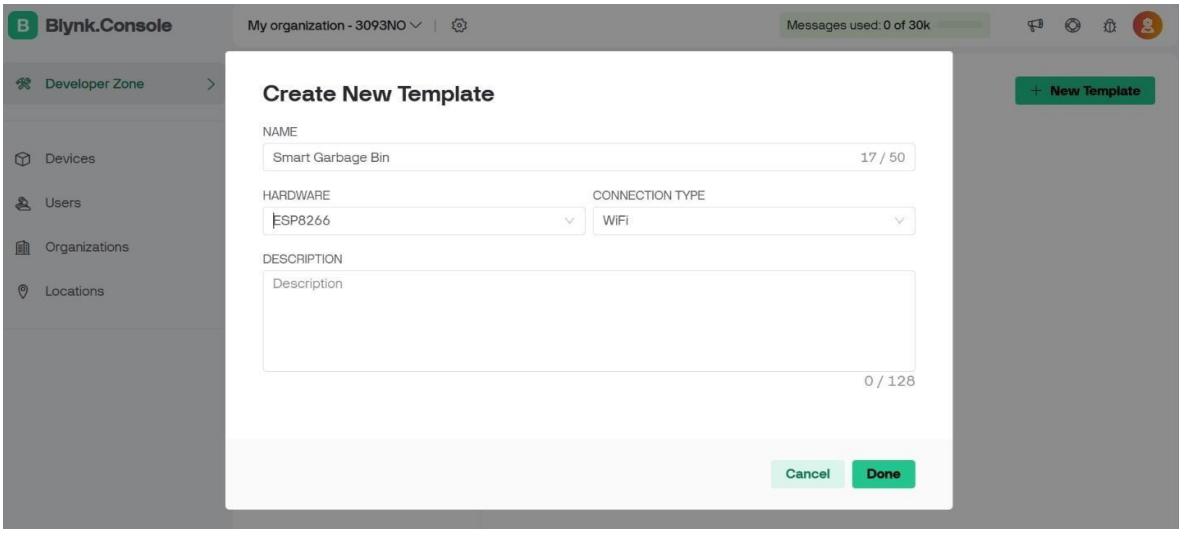

Fig 5.3: New Template Creation

Step 4: Create required data streams by selecting the virtual pin.

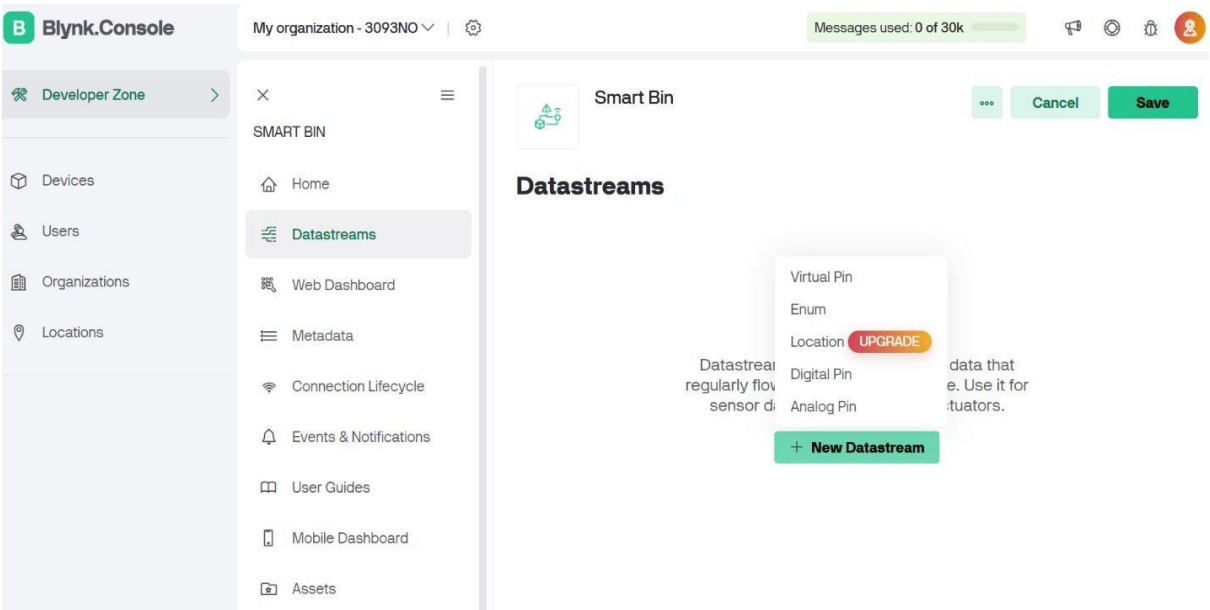

Fig 5.4: Selecting Virtual Pin

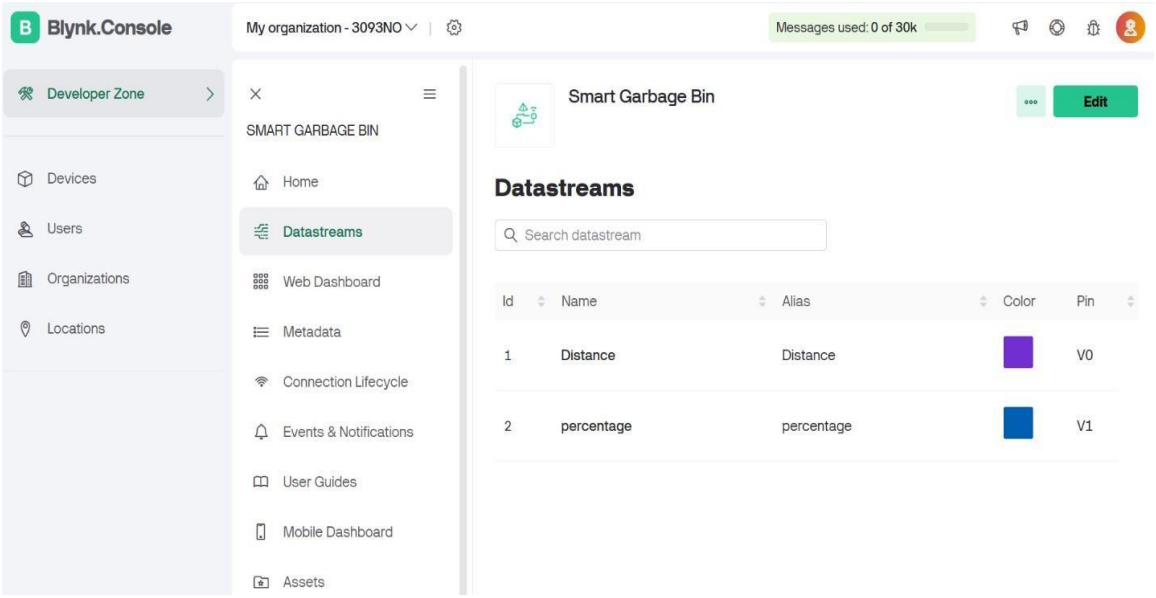

Fig 5.5: Data streams

Step 5: Create Web Dashboard.

After that, click and drag Guage widgets according to the number of relays.

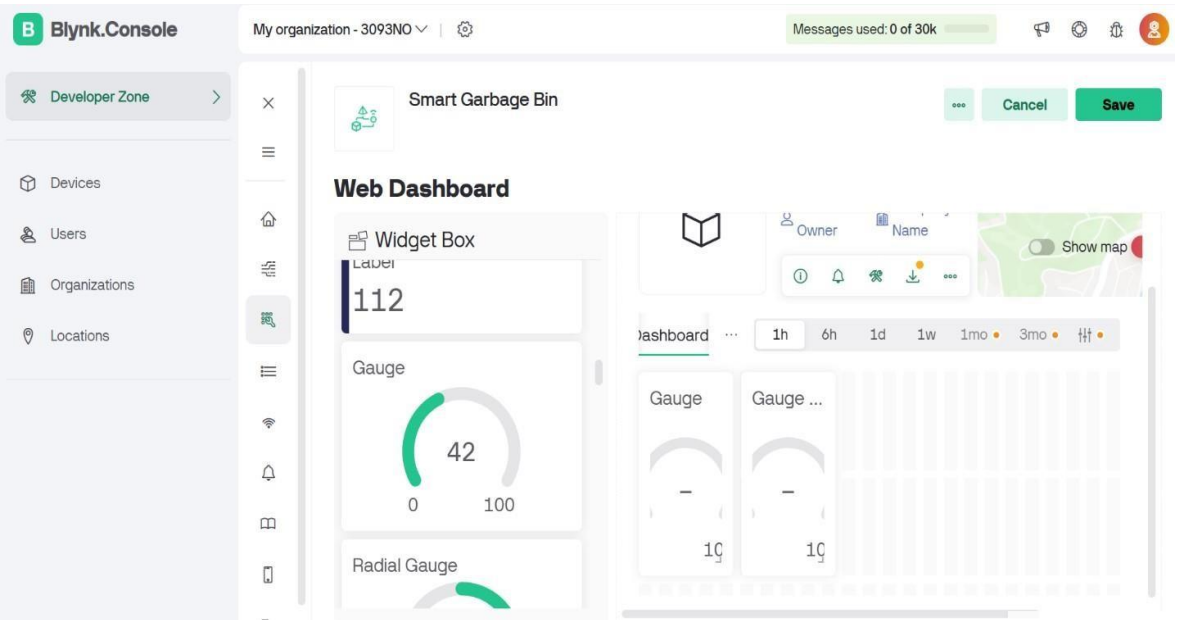

Fig 5.6: Adding Widgets

Step 6: Click on “Save”.

Step 7: Then create a new device “Smart Garbage Bin”.

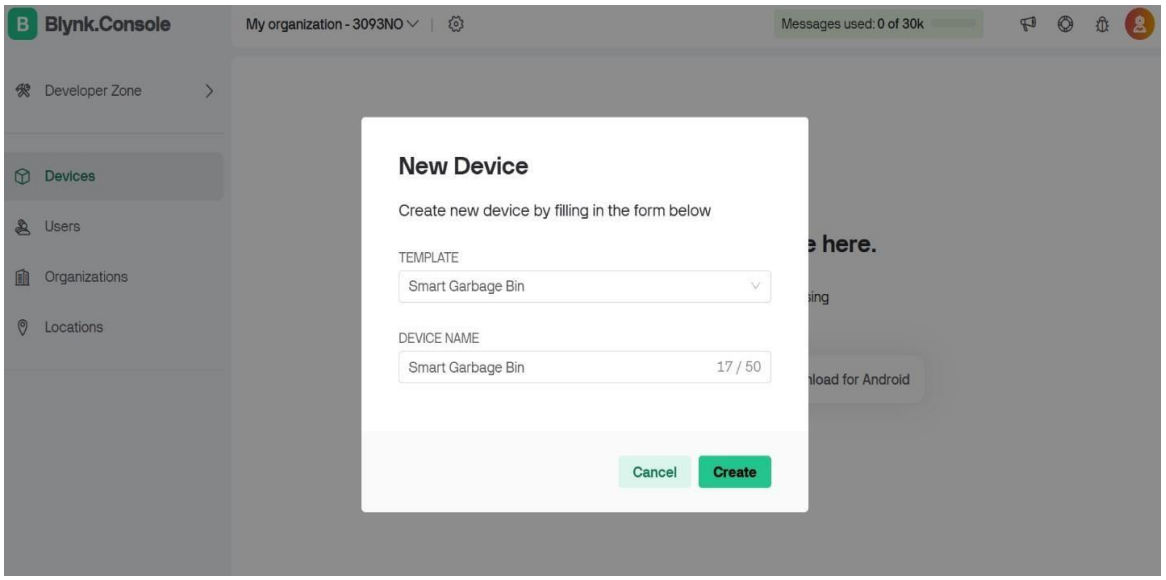

Fig 5.7: New Device Creation

Step 8: An Auth Token is sent to the registered mail or created new device.

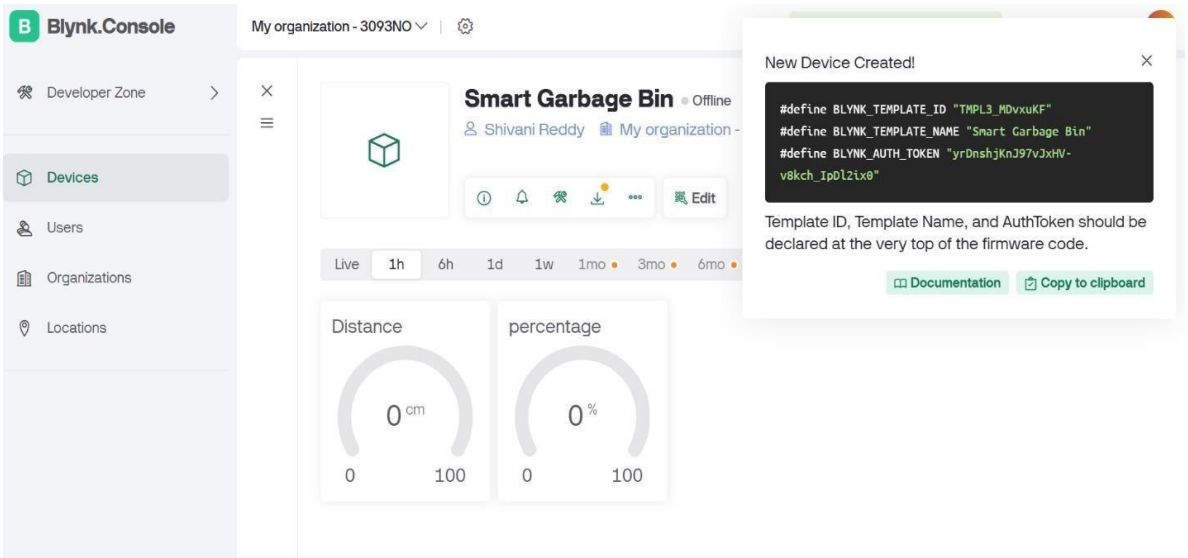

Fig 5.8: Auth Token

HARDWARE SETUP:

Step 1: Give the connections according to the following table.

Ultrasonic Sensor	Node MCU Pin (ESP8266)	Description
VCC	3V3	Connect to the 3.3V output of the Node MCU
GND	GND	Connect to the GND pin of the Node MCU
TRIG	D0	Connect to digital pin D0 (GPIO16)
ECHO	D1	Connect to digital pin D1 (GPIO5).

Step 2: The ultrasonic sensor is mounted at the top of the garbage bin, which measures the distance between sensor and the garbage.

Step 3: The NodeMCU receives the distance data from the ultrasonic sensor via its GPIO pins.

Step 4: Define a threshold value and maximum distance in the code.

Step 5: The maximum distance (i.e., the height of the bin) and the measured distance are used to calculate the garbage fill percentage:

Fill Percentage = (1−Maximum Distance / Measured Distance) ×100 **Step 6:** Now connect the Node MCU to the PC and open Arduino IDE.

Step 7: As all the required libraries are already installed, update the Wi-Fi credentials in the code.

- Add your SSID and Password in the code as 'YourNetworkName' and 'YourPassword'.
- Paste the **Authentication Token** that you received in your email at 'YourAuthToken'.

Step 8: Check whether you have chosen the right port and Board before uploading the code.

Step 9: Once the code is compiled and uploaded, the garbage level will be displayed in the serial monitor.

Step 10: As the NodeMCU connects to a Wi-Fi network and the fill percentage is sent to the Blynk Application which displays the fill level in real time, enabling users to monitor the bin remotely.

Step 11: If the calculated fill percentage exceeds the threshold, an alert is triggered.

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM ARCHITECTURE

The system architecture of a **Smart Garbage Monitoring System** using IoT integrates sensors, microcontrollers, and cloud-based applications for efficient waste management. Ultrasonic sensors installed on garbage bins measure waste levels and send data to an IoT-enabled microcontroller, such as Node MCU or Arduino with Wi-Fi modules.

The microcontroller processes this data and transmits it to a cloud platform via the Internet. Users access real-time garbage level updates through a mobile or web application. Notifications are sent when bins are full, and data analytics can optimize waste collection schedules. This architecture ensures efficient monitoring, timely waste disposal, and supports smart city initiatives.

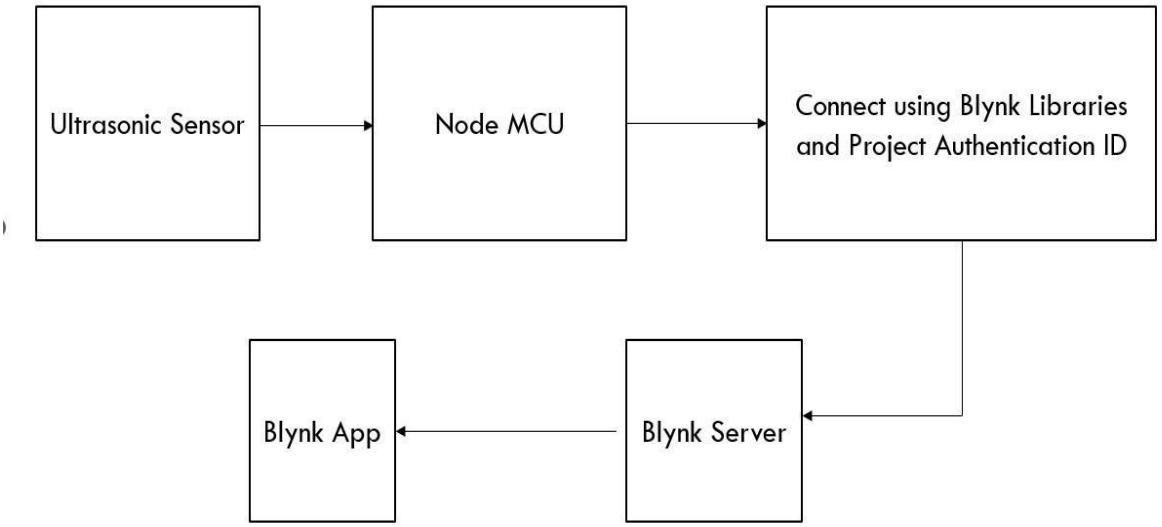

Fig 6.1: System Architecture

6.2 DATA FLOW DESIGN

A Data Flow Diagram (DFD) is a graphical representation of the flow of data within a system or process. It is a modelling technique that shows how data moves through processes and how it is stored, transformed, or exchanged within a system. DFDs are often used in system analysis and design to visualize and understand the data processing and flow of information in a structured manner.

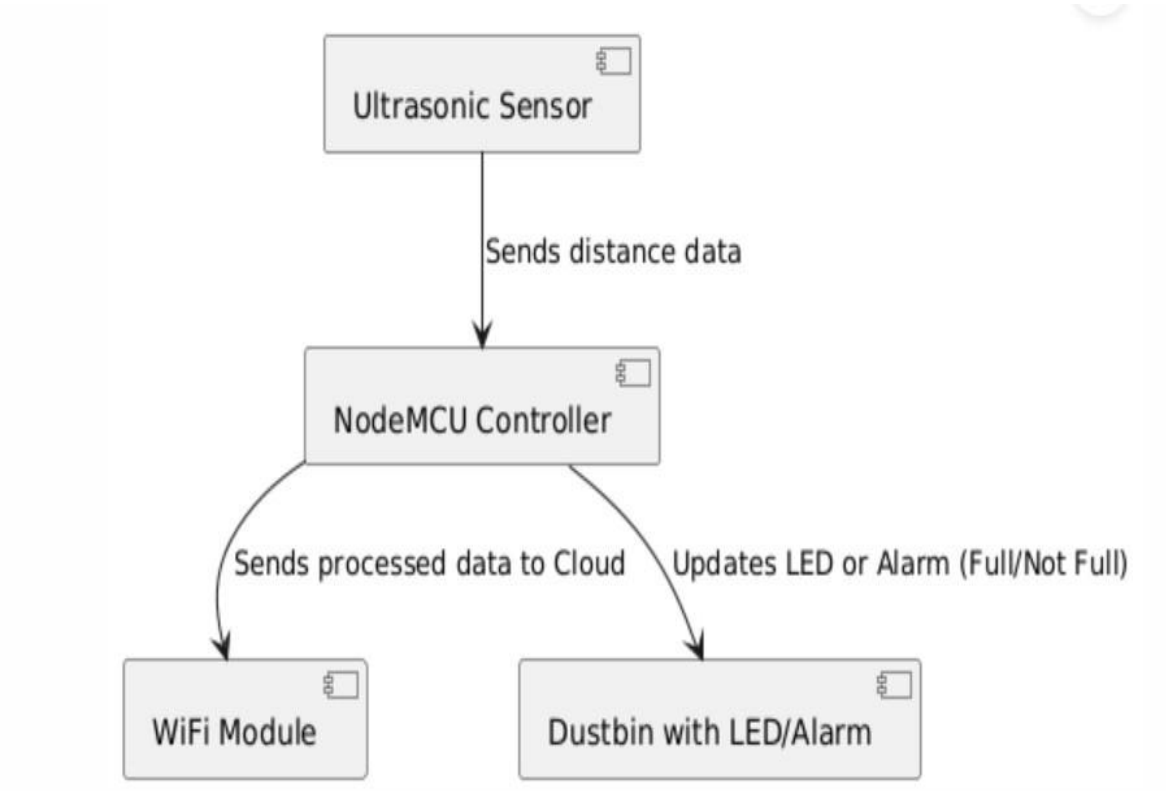

Fig 6.2.1: Data Flow Diagram for Garbage Bin

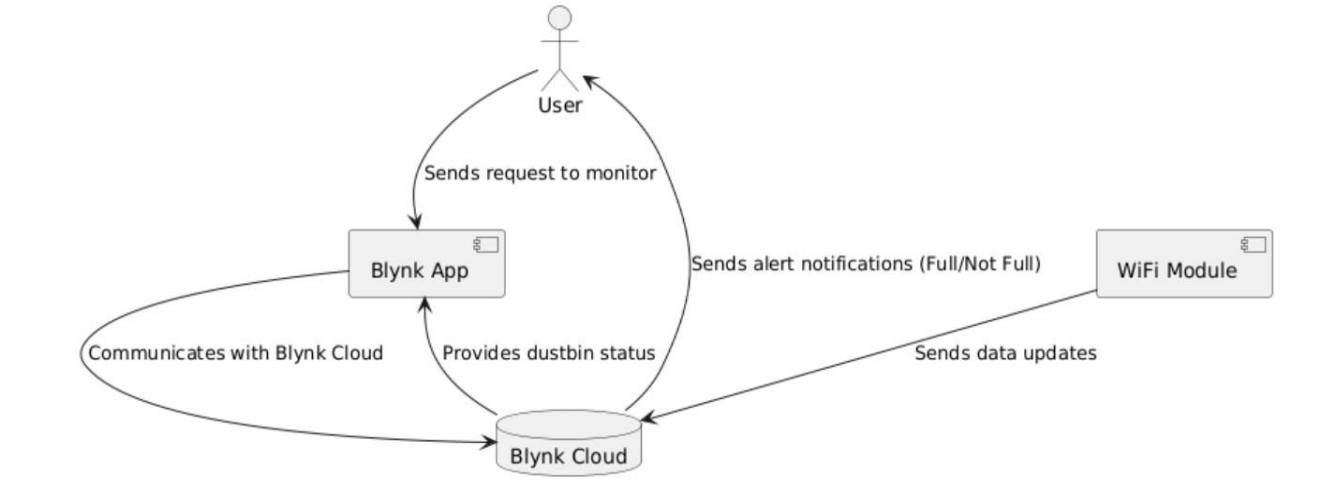

Fig 6.2.2: Data Flow Diagram for Blynk application

6.3 UML DESIGN

Unified Modelling Language (UML) is a standardized modelling language in the field of software engineering. It provides a way to visualize a system's design through a set of diagrams. UML diagrams help software developers and system architects communicate and understand the structure and behaviour of a system. In UML, the diagrams can be broadly categorized into two main types: structural diagrams and behavioural diagrams.

1. Structural Diagram

Structural diagrams in UML focus on representing the static structure of a system. They showcase the components that make up the system and their relationships. The following are different structural diagrams:

1. Class diagram
2. Object diagram

3. Component diagram

4. Deployment diagram

2. Behavioural Diagram

Behavioural diagrams primarily capture the dynamic facet of the system. Dynamic aspects can be more delineated because of the dynamical or moving elements of a system.

UML has the subsequent 5 varieties of behavioural diagrams.

They are:

1. Use case diagram
2. Sequence diagram
3. Collaboration diagram
4. State chart diagram
5. Activity diagram

6.3.1 Use Case Diagram

Use Case Diagrams in UML describe interactions between a system and external entities known as actors. Use cases represent specific functionalities or scenarios that the system provides to its users.

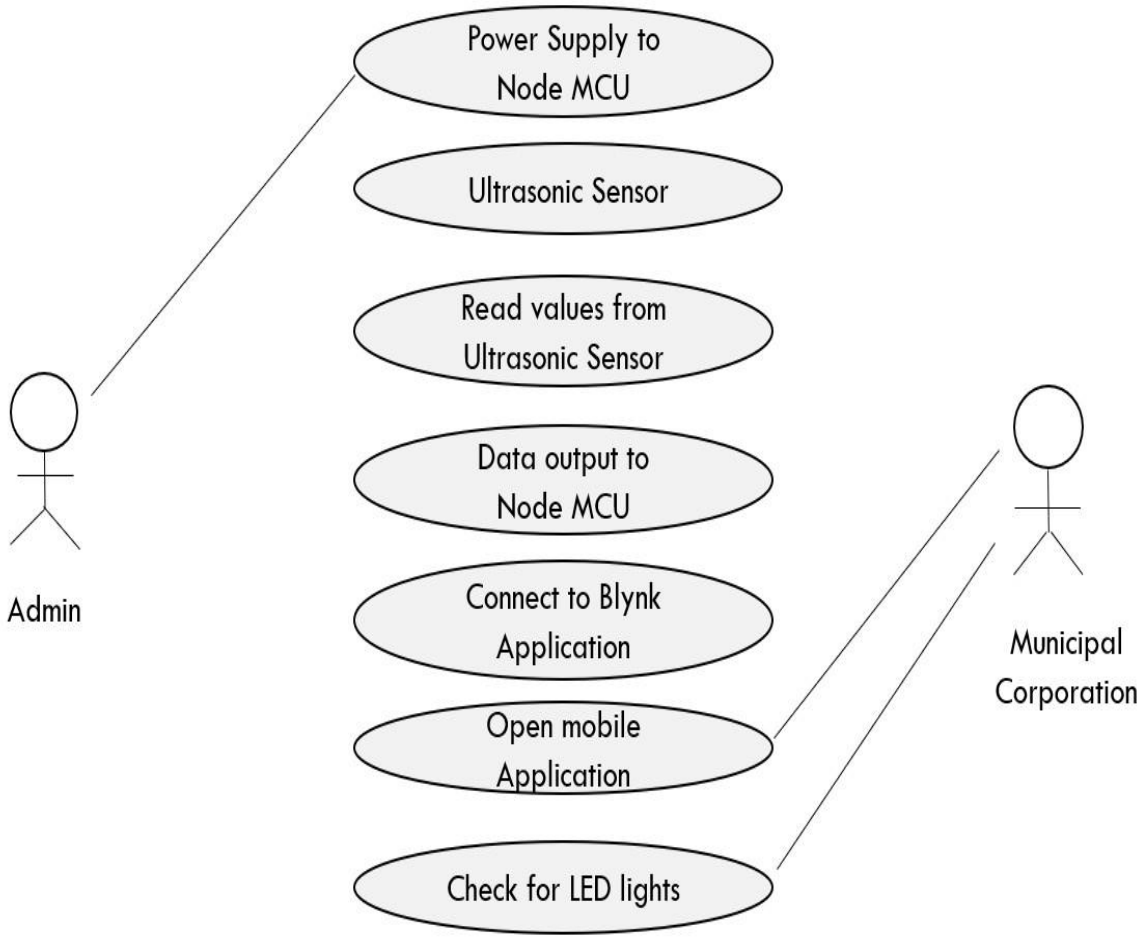

Fig 6.3.1: Use Case Diagram

6.3.2 Class Diagram

The Class Diagram in UML illustrates the static structure of a system, detailing classes, attributes, methods, and their relationships. Rectangles represent classes, and lines indicate associations, dependencies, and inheritances.

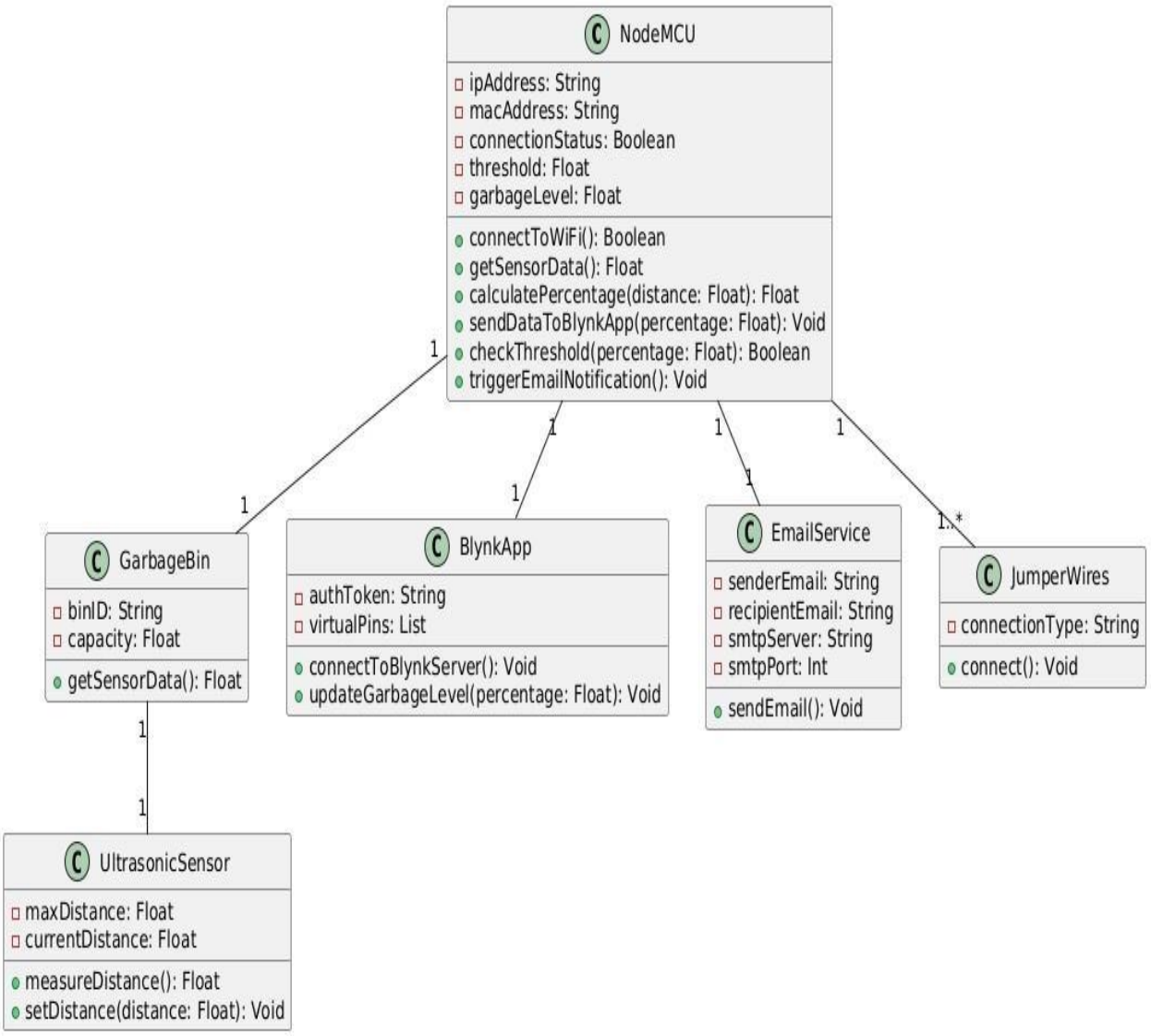

Fig 6.3.2: Class Diagram

6.3.3 Activity Diagram

Activity Diagrams represent the flow of activities within a process or workflow. They focus on actions, decisions, and control flows, providing a high-level view of the dynamic aspects of a system.

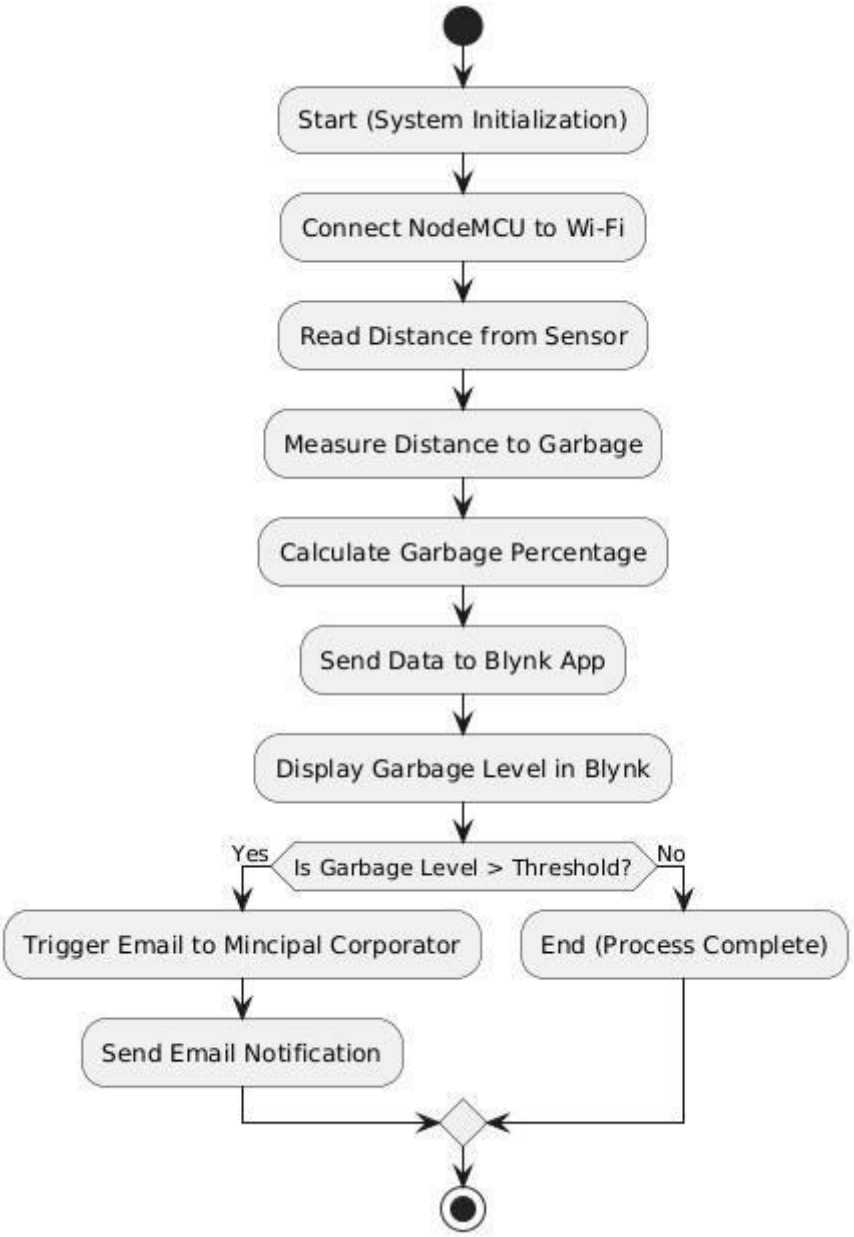

Fig 6.3.3: Activity Diagram

6.3.4 Sequence Diagram

A sequence diagram merely depicts interaction between objects in an exceedingly serial order i.e., the order during which these interactions turn up. We will conjointly use terms event diagrams or event eventualities to consult with a sequence diagram. These diagrams are widely employed by businessmen and software package developers to document and perceive needs for brand new and existing systems.

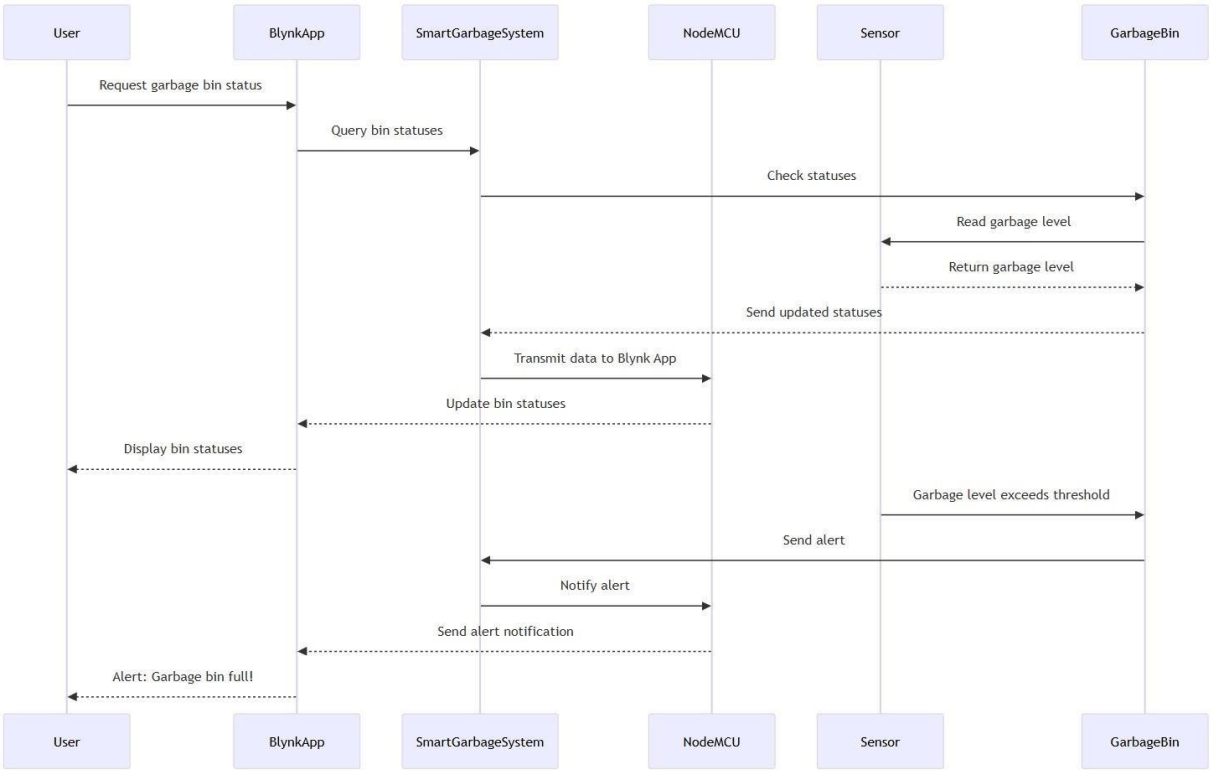

Fig 6.3.4: Sequence Diagram

CHAPTER 7 IMPLEMENTATION

The system comprises a Node MCU (ESP8266) microcontroller, an ultrasonic sensor (HC-SR04), and the Blynk mobile application. The ultrasonic sensor, mounted on the lid of a garbage bin, measures the distance between the sensor and the waste level to determine how full the bin is. Node MCU processes this data and sends real-time updates to the Blynk app via Wi-Fi. The Blynk app displays the garbage level and sends notifications when the bin is full, ensuring timely waste disposal. If the waste level exceeds a predefined threshold, the system triggers an email alert to municipal authorities for prompt action.

The hardware setup involves connecting the ultrasonic sensor to Node MCU using jumper wires for power and signal transmission, while software implementation requires the Arduino IDE for programming. The Blynk library and ESP8266WiFi library are used to integrate the system with the app and establish Wi-Fi connectivity. The garbage level data is periodically measured and transmitted to the app, providing real-time monitoring and reducing manual inspection efforts. The system is cost-effective, user-friendly, and scalable, making it suitable for residential, commercial, and public spaces. By preventing bin overflow, it enhances hygiene and contributes to a cleaner environment.

7.1 INSTALLATIONS AND LIBRARIES USED

- Install the latest version of the Arduino IDE.
- Add the ESP8266 board package through the Board Manager.
- Include the "ESP8266WiFi" library.
- Install the Blynk library “BlynkSimpleEsp8266.h” from the Arduino Library Manager.
- Go to code > Download ZIP.
- Open Arduino IDE > Sketch > Add .ZIP Library.
- Select the ZIP files that you just downloaded.

7.2 CODE

```
#define BLYNK_TEMPLATE_ID "TMPL3JEYdy05x"
#define BLYNK_TEMPLATE_NAME "Smart Garbage bin"
#define BLYNK_AUTH_TOKEN "xmzbJucPWZ1pXOWg0uitgOPZlCIs03k4"
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h> // Wi-Fi and Blynk configuration  char ssid[]
= "YourNetworkName”;          // Replace with your Wi-Fi SSID char pass[] =
"YourPassword";      // Replace with your Wi-Fi password
// Replace with your Blynk Auth     Token char auth[]
="YourAuthToken”;

// Ultrasonic Sensor pins
#define TRIG_PIN D0
#define ECHO_PIN D1
// Bin dimensions
const int binHeight = 30; // Bin height in cm (adjust based on actual bin height)

// Function to notify when the bin is full void
notifyWhenFull(int percentageFull) {   if
(percentageFull >= 85) {
    Serial.println("Garbage bin is full!");
    // Trigger a Blynk event notification when the bin is full
    Blynk.logEvent("smart_bin", "Garbage bin is full!");
  }  }  void
setup() {
  // Initialize Serial Monitor
```

```
  Serial.begin(19200);

  // Set up Blynk
  Blynk.begin(auth, ssid, pass);

  // Configure ultrasonic sensor pins
pinMode(TRIG_PIN, OUTPUT);   pinMode(ECHO_PIN,
INPUT);
}          void
loop() {
  Blynk.run(); // Run Blynk

  // Measure distance and calculate fullness percentage   int
distance = measureDistance();       int percentageFull =
calculatePercentageFull(distance);
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm | Fullness: ");
  Serial.print(percentageFull);
  Serial.println("%");

  // Send distance and fullness percentage to Blynk virtual pins
  Blynk.virtualWrite(V0, distance);        // Send distance to V0
  Blynk.virtualWrite(V1, percentageFull);   // Send fullness percentage to V1

  // Check if bin is full and send notification
notifyWhenFull(percentageFull);   delay(1000);
// Update every minute
}
```

```
// Function to measure distance using ultrasonic sensor
int measureDistance() {    digitalWrite(TRIG_PIN,
LOW);   delayMicroseconds(2);
digitalWrite(TRIG_PIN, HIGH);
delayMicroseconds(10);
digitalWrite(TRIG_PIN,   LOW);           long
duration = pulseIn(ECHO_PIN, HIGH);
int distance = duration * 0.034 / 2; // Convert time to distance in cm    return
distance;
}

// Function to calculate fullness percentage  int calculatePercentageFull(int distance)
{    int percentageFull = map(distance, binHeight, 0, 0, 100);  // Map distance to 0-
100%   percentageFull = constrain(percentageFull, 0, 100);        // Limit to 0-100%
return percentageFull;
}
```

CHAPTER 8

RESULTS

This section describes step-by-step the system analysis and the results obtained. The system was designed and simulated in proteus and a working prototype was developed using the different hardware components with data being sent to the mobile application.

CODE OUTPUT:

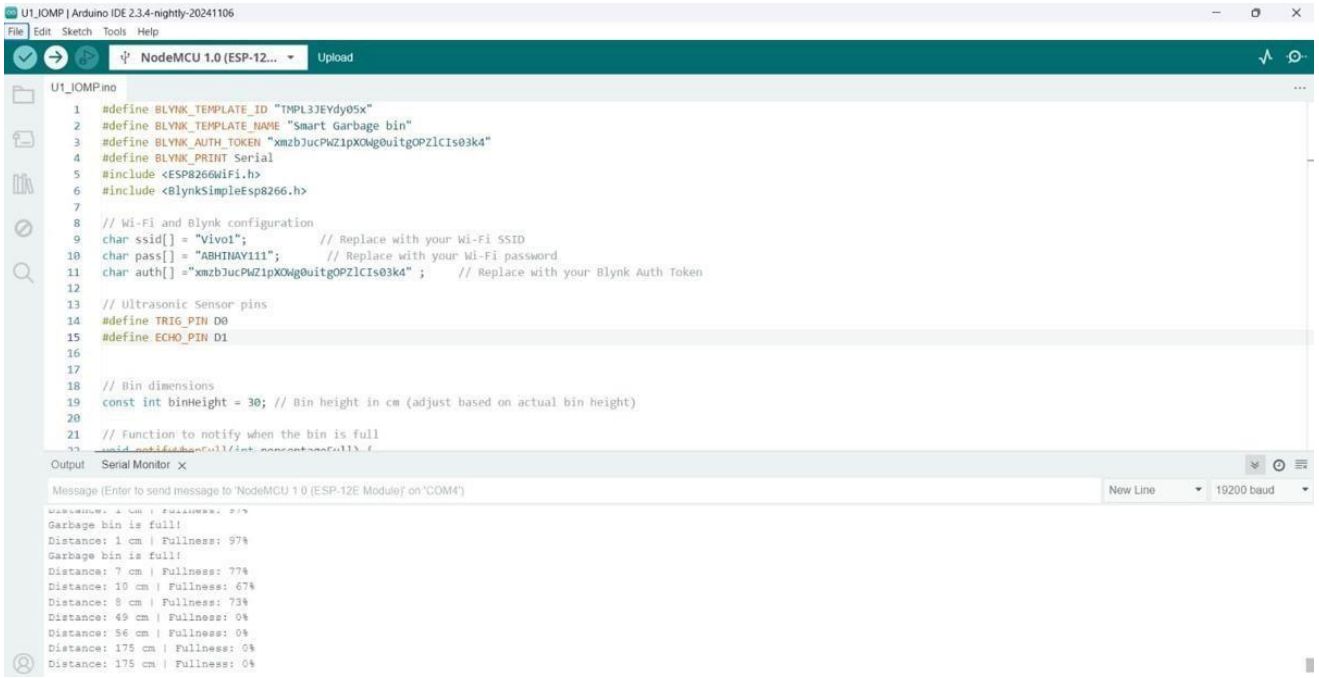

Fig 8.1: Code Output in Serial Monitor

APPLICATION INTERFACE:

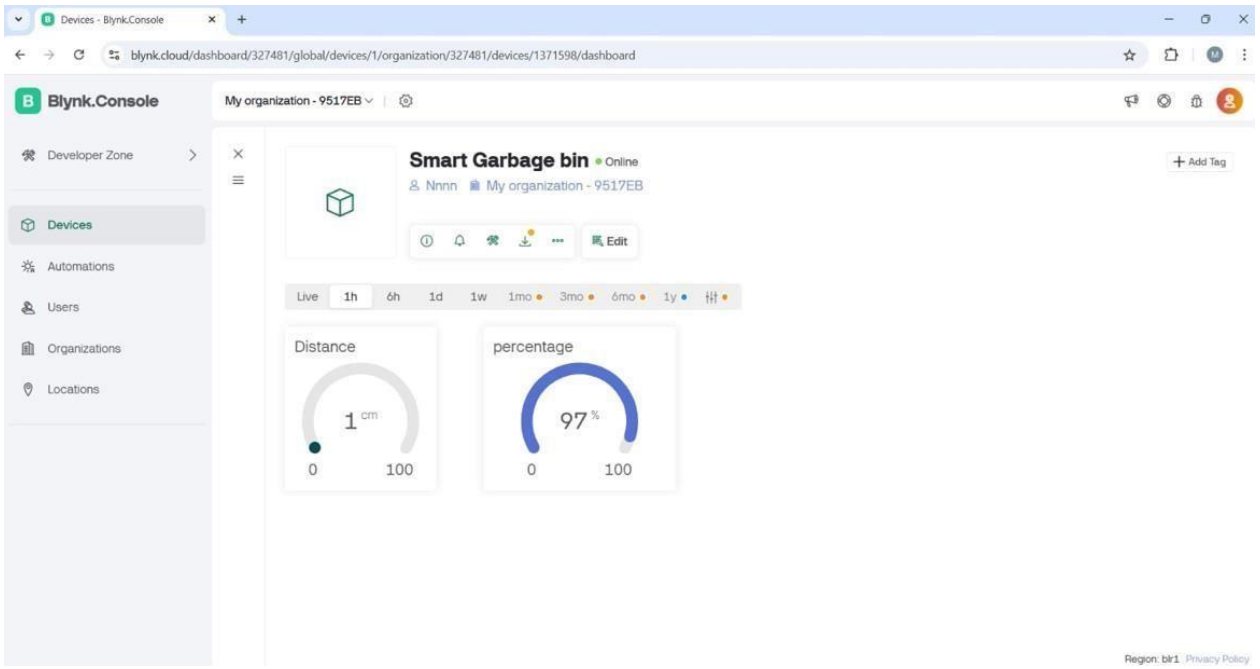

Fig 8.2: Blynk Application Interface

EMAIL ALERT:

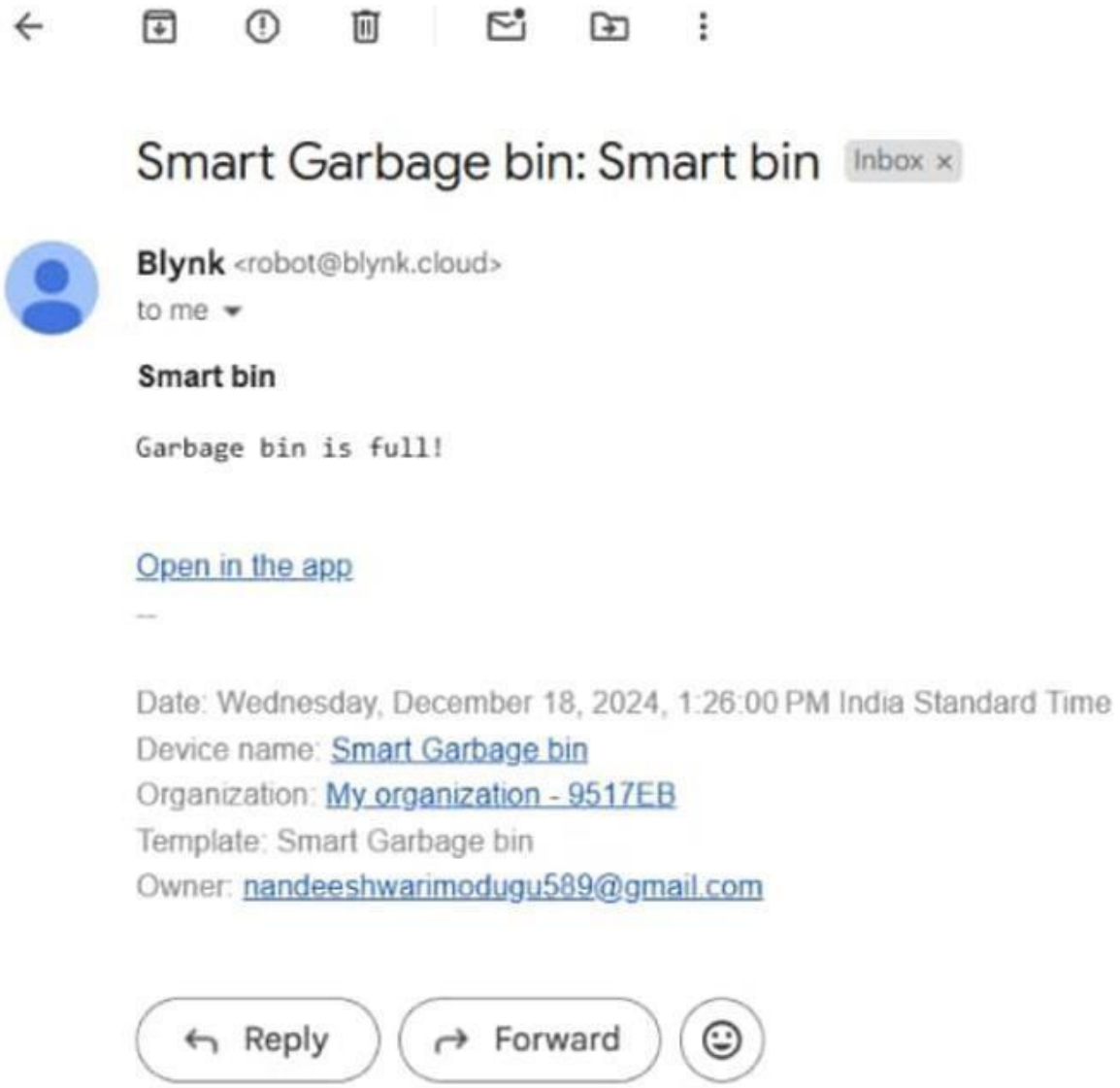

Fig 8.3: Alert Message to Mail

CHAPTER 9

FUTURE SCOPE

While the Smart Garbage Monitoring System meets the defined scope and requirements, there is room for future enhancements, including:

- Integration with IoT Platforms.
- Multi-Sensor Integration.
- Solar-Powered Systems.
- Mobile App Enhancements.
- Scalability for Larger Networks.
- Public Awareness Integration.

The system in this project can also be expanded to monitor other environmental parameters, such as odour levels or toxic gases, ensuring better public health and safety. With increasing urbanization and the demand for smart cities, this technology has the potential to play a critical role in building cleaner, more efficient urban environments.

CHAPTER 10

CONCLUSION

The Smart Garbage Monitoring System using Node MCU and the Blynk app offers a practical and innovative solution to modern waste management challenges. By integrating an ultrasonic sensor with a Node MCU microcontroller, the system is capable of accurately measuring the waste levels in garbage bins and transmitting real-time data to a user-friendly mobile application. This ensures that the municipal authorities or waste management teams are alerted as soon as a bin is full, allowing for timely collection and preventing overflows. The use of the Blynk app adds another layer of convenience, enabling users to monitor waste levels remotely and receive notifications directly to their smartphones.

This system's implementation enhances operational efficiency by reducing the need for manual inspections of garbage bins, saving both time and resources. Moreover, it contributes to environmental sustainability by ensuring that waste is managed more effectively and efficiently, reducing the risk of littering and pollution in public spaces. The integration of Internet of Things (IoT) technology provides scalability and flexibility, allowing the system to be expanded and customized for use in various settings, including residential, commercial, and public areas.

Furthermore, the system's ability to send email alerts to municipal corporations paves the way for smart cities where waste collection is optimized based on actual data rather than predefined schedules. Looking ahead, the system can be further improved by integrating advanced features such as machine learning for predicting garbage accumulation patterns and incorporating renewable energy sources to make the system more sustainable. Overall, the Smart Garbage Monitoring System not only streamlines waste management but also contributes to the creation of smarter, cleaner, and more efficient urban environments.

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